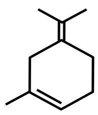
Single choice Question with one option correct	**Q.01 to Q.10**
Multiple choice Question with one or more than one may be correct	**Q.11 to Q.30**
Numerical Based Question (no negative marking)	**Q.31 to Q.50**
Stem Question	**Q.51 to Q.54**

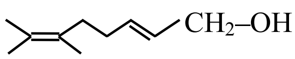
DPP Syllabus : Hydrocarbons, Haloalkanes, Haloarenes, Alcohols and Ethers (Part-1 & 2) Reduction, Oxidation & Hydrolysis reactions

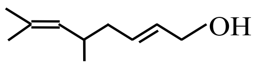
1. Geraniol, $C_{10}H_{18}O$, a terpene found in rose oil, adds two moles of bromine to form a tetrabromide, $C_{10}H_{18}OBr_4$. It can be oxidized to a ten-carbon aldehyde or to a ten-carbon carboxylic acid. Upon vigorous oxidation, geraniol yields:

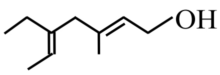
$CH_3-\underset{\underset{O}{\|}}{C}-CH_3 \quad CH_3-\underset{\underset{O}{\|}}{C}-CH_2-CH_2-\underset{\underset{O}{\|}}{C}-OH \quad HO-\underset{\underset{O}{\|}}{C}-\underset{\underset{O}{\|}}{C}-OH$

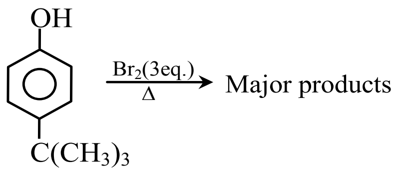
Structure of geraniol will be:

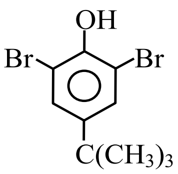
(A) 

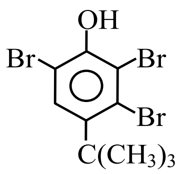
(B) 

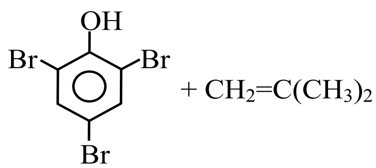
(C) 

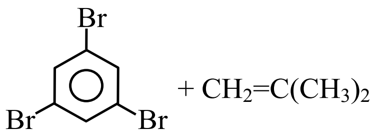
(D) 

2. 

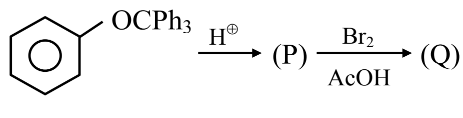
(A) 

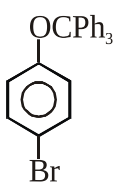
(B) 

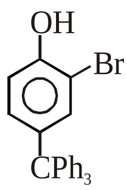
(C) 

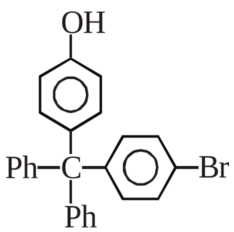
(D) 

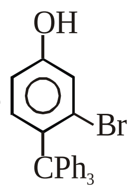
3. The major product (Q) obtained in the following reaction is :

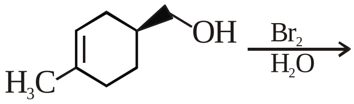


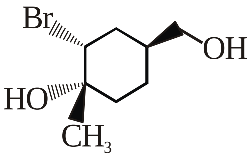
(A) 

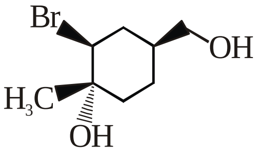
(B) 

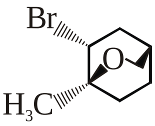
(C) 

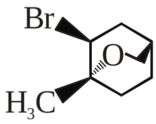
(D) 

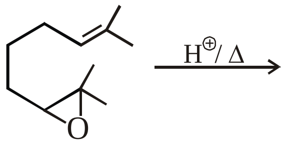
4.  (P) major product

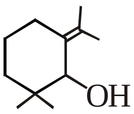
(A) 

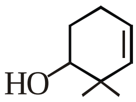
(B) 

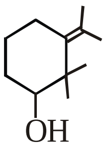
(C) 

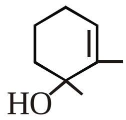
(D) 

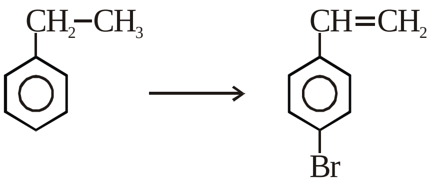
5.  Product

(A) 

(B) 

(C) 

(D) 

6. 

Select the best series of reagent to bring out the above conversion.

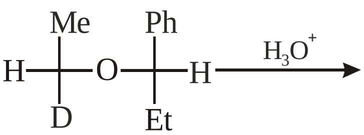
(A) (i) NBS/Δ, (ii) CH_3ONa/CH_3OH, Δ, (iii) Fe/Br_2

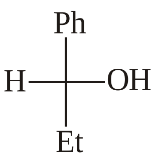
(B) (i) Fe/Br_2, (ii) NBS/Δ, (iii) CH_3ONa/CH_3OH, Δ

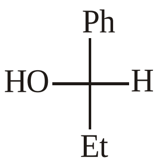
(C) (i) $Br_2/h\nu$, (ii) CH_3ONa/CH_3OH, Δ, (iii) HBr

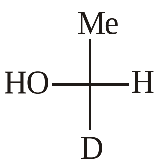
(D) (i) $Br_2/h\nu$, (ii) NBS/Δ, (iii) CH_3ONa/CH_3OH, Δ

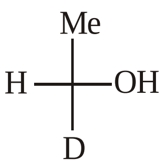
7. Which of the following alcohol is not present in the mixture containing major products?

 Products mixture.

(A) 

(B) 

(C) 

(D) 

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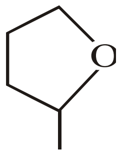
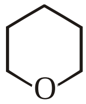
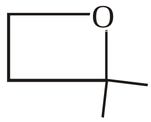
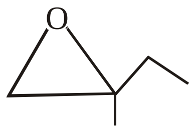
8. $\underset{\substack{(X)\\\text{Optically}\\\text{active}}}{C_5H_{10}O} \xrightarrow[\text{excess}]{HI} \underset{\substack{(Y)\\\text{Optically}\\\text{active}}}{C_5H_{10}I_2} \xrightarrow{KOH(alc)} \underset{\substack{(Z)\\\text{Conjugated}\\\text{diene}}}{C_5H_8}$

Which is the best possible structure of X from the given compounds ?

(A) 
(B) 
(C) 
(D) 

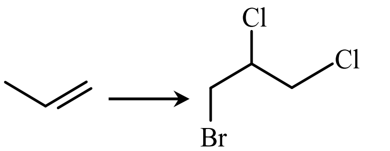
9. But-1-ene is obtained as a major product in the given reaction, when ?

(A) G will be F

(B) G will be $-\overset{\oplus}{N}R_3$

(C) Base will be $CH_3-\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}-OK$

(D) All of these

10. 

Above conversion can be brought about by ?

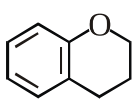
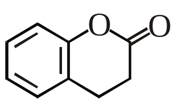
(A) (i) HBr ; (ii) Cl_2/CCl_4

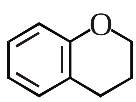
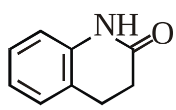
(B) (i) $Cl_2/h\nu$; (ii) NBS

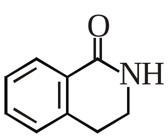
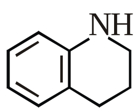
(C) (i) NBS ; (ii) Cl_2/CCl_4

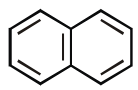
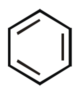
(D) (i) Cl_2/CCl_4 ; (ii) BrCl / CCl_4

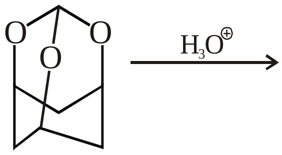
11. Which of the following is/are correct order of reactivity towards electrophilic subsitution reaction with Br_2/Fe?

(A)  > 

(B)  > 

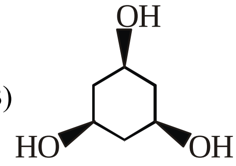
(C)  > 

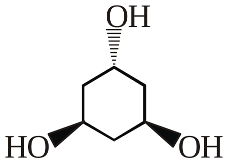
(D)  > 

12. 

The product obtained in following reaction is/are?

(A) HCHO

(B) 

(C) 

(D) HCOOH

13. 

(P)

(Q)

Which of the following statement(s) is/are correct for above pair of reactions?

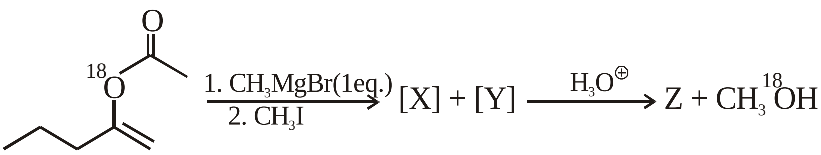
(A) Reactions are stereospecific and stereoselective.

(B) Both P and Q form anhydride on heating.

(C) (Y) is meso-tartaric acid while (X) is racemic mixture.

(D) The most stable conformation of (Y) is Gauche form.

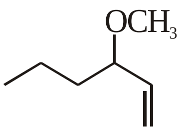
14. Observe the following reaction, and answer as per given reaction only :-



Hint: X is an unsaturated ether

If Y gives yellow precipitate with NaOI then which of the following statement(s) is/are correct?

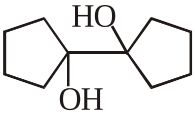
(A) [Z] on reaction with hydroxyl amine gives two oximes.

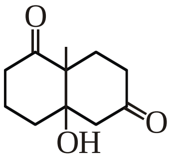
(B) [X] is 

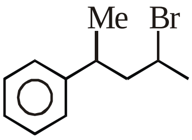
(C) [Z] gives butanoic acid on reaction with bleaching powder followed by acidification.

(D) [Z] is ketone.

15. Which of the following is/are correct for given reactions ?

(I)  $\xrightarrow[\Delta]{\text{Conc. } H_3PO_4}$ Product

(II)  $\xrightarrow{\text{dil. NaOH}/\Delta}$ Product

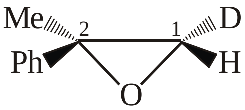
(III)  $\xrightarrow{\text{Fe},Br_2}$ Product

(A) Reaction I has a spirocyclic compound as a product.

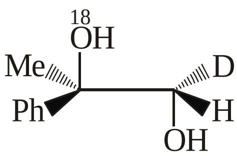
(B) Reaction II has a carbanion intermediate.

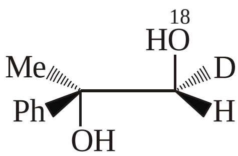
(C) Reaction I & II both involve dehydration.

(D) Reaction I, II, III all are elimination reactions.

16.  $\xrightarrow{H^{\oplus}/H_2O^{18}}$ (Product)

The correct observations about the above mentioned reaction are :

(A)  (Major product)

(B)  (Major product)

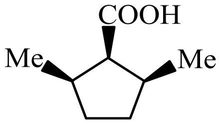
(C) Reactant has 1S, 2S configuration.

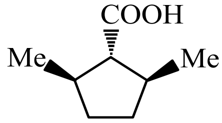
(D) Major product has 1S, 2R configuration.

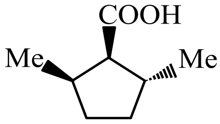
17. 2,5-Dimethyl-1,1-cyclopentanedicarboxylic acid (I) can be prepared as two optically inactive substances (P and Q) of different melting points.

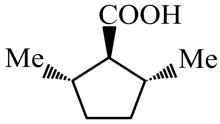
Upon heating P yields two 2,5-Dimethylcyclopentanecarboxylic acids (II) and Q yields only one.

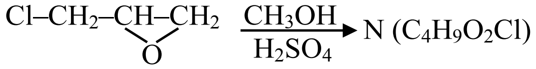
Identify products of P & Q:

(A) Product of Q is 

(B) Product of Q is 

(C) Product of P is 

(D) Product of P is 

18. 

$N + NaOCl \longrightarrow CHCl_3 + Z\ (C_3H_6O_3)$

$N + NaOH(aq.) \longrightarrow P\ (C_4H_8O_2)$

(A) N is $ClCH_2CH(OH)CH_2OCH_3$

(B) Z is $CH_3–O–CH_2COOH$

(C) P has both cyclic & open chain ether.

(D) The conversion of N to P is an intramolecular electrophilic substitution reaction.

19. Styrene $\xrightarrow[\text{2. NaNH}_2\ \text{(excess)}]{\text{1. Br}_2}$ M $\xrightarrow{CH_3I}$ N $\xrightarrow[H^+/H_2O]{Hg^{2+}}$ P $\xrightarrow{NaBH_4}$ Q $\xrightarrow[\text{heat}]{H_2SO_4}$ R $\xrightarrow[EtOH]{Br_2}$ S

Correct statement(s) for the above reactions is/are :

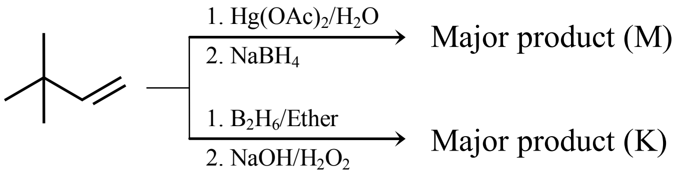
(A) Product (N) is 1-phenyl propyne.

(B) Product (Q) gives red colour with ceric ammonium nitrate.

(C) Major product (S) will be $Ph–\underset{Br}{CH}–\underset{OC_2H_5}{CH}–CH_3$

(D) Product (P) decolourises bromine solution.

20. Correct about major products M & K is/are :



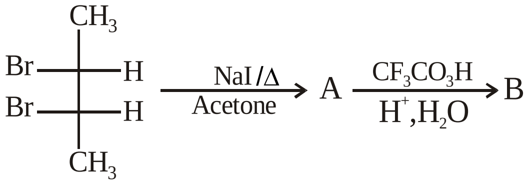
(A) M & K are stereoisomers of each other.

(B) M & K are structure isomers of each other.

(C) K is primary and M is secondary alcohol.

(D) M is ketone and K is aldehyde.

21. The correct statements for the given reaction is :



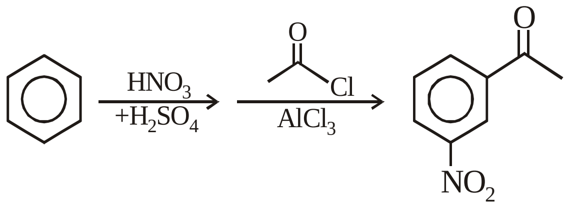
(A) B is optically inactive due to external compensation.

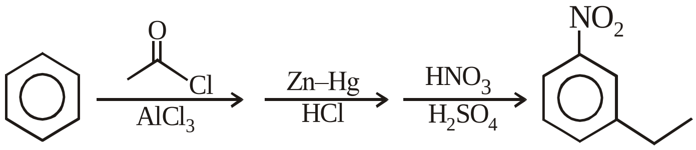
(B) B is optically inactive due to internal compensation.

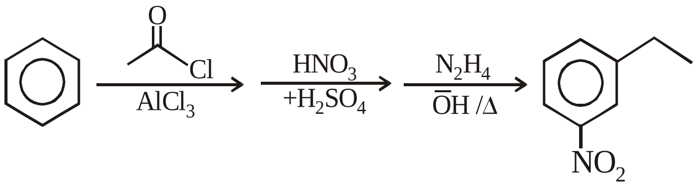
(C) A is predominantly trans-alkene.

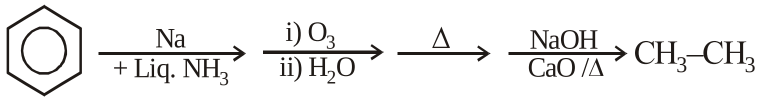
(D) B does not have chiral centres.

22. In the given reactions major products are shown, except ?

(A) 

(B) 

(C) 

(D) 

23. Sequence of reagents used to convert cis-stilbene into trans-stilbene 'will be' :

(A) (i) Br_2 (ii) $NaNH_2$(excess), Δ(iii) Na,$NH_3(\ell)$

(B) (i) HBr (ii) alc. KOH, Δ

(C) (i) dil, H_2SO_4 (ii) conc. H_3PO_4

(D) (i) Br_2, CCl_4 (ii) Zn dust, Δ

24. In which of the following reaction the resolvable products is / are obtained

(A) $CH_3—C \equiv C—CH_3 \xrightarrow[(2)\ Br_2\ /CCl_4]{(1)\ H_2/Pd\text{-}BaSO_4}$

(B) $C_2H_5—C \equiv C—C_2H_5 \xrightarrow[(2)\ Br_2\ /CCl_4]{(1)\ Li\ /\ NH_3}$

(C) $CH_3—C \equiv C—CH_3 \xrightarrow[(2)\ dil.\ KMnO_4]{(1)\ H_2/Pd\text{-}BaSO_4}$

(D) $CH_3CH{=}O + HCN \longrightarrow$

25. Product(s) obtained during complete hydrolysis of

$CH_3 – CH = CH – O – CH_2 – CH_3 \xrightarrow{H_3O^{\oplus}}$ Product (s)

(A) $CH_3 – CH_2 – CH_2 – OH$

(B) $CH_3 – CH = O$

(C) $CH_3 – CH_2 – OH$

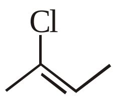
(D) $CH_3 – CH_2 – CH = O$

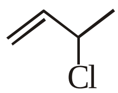
26.  $\xrightarrow{NBS} A \xrightarrow{(aq.)\ NaOH} B \xrightarrow[\Delta]{H^+} C \xrightarrow{HCl\ /\ CCl_4,\ (1\ eq.)}$ Products

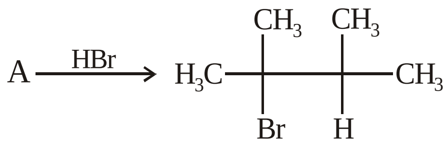
Final possible products is/are:

(A) 

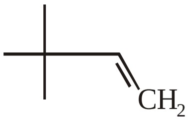
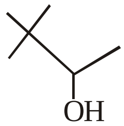
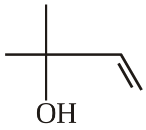
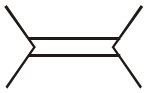
(B) 

(C) 

(D) 

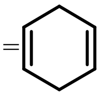
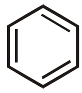
27. 

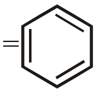
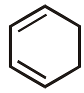
The reactant A can be :

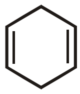
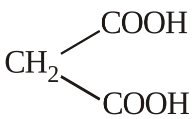
(A)  (B)  (C)  (D) 

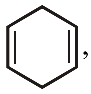
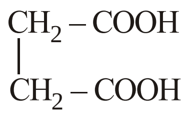
28. $\underset{\text{(Hydrocarbon)}}{P} \xrightarrow{Na / liq.\ NH_3} Q \xrightarrow[(ii)\ H_2O]{(i)\ O_3} 2R \xrightarrow{\Delta} 2CH_3COOH + 2CO_2$

Identify the unknowns :

(A) Q =  , P = 

(B) P =  , Q = 

(C) Q =  , R = 

(D) Q =  , R = 

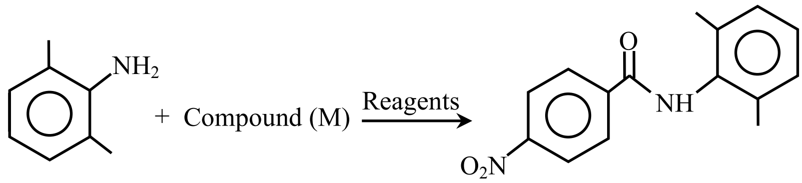
29. An optically inactive alcohol (A) $C_6H_{12}O$ is oxidized by MnO_2 to produce optically inactive carbonyl compound while reduction of (A) by H_2/Ni produces optically active compound. Possible structure(s) of alcohol is/are

(A) Hex-2-ene-1- ol

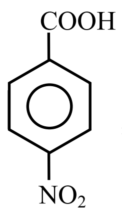
(B) Hex-3-ene-2- ol

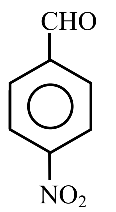
(C) 2-Methyl pent-2 - ene-1-ol

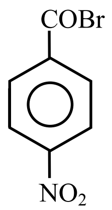
(D) 3-Methyl pent-2 - ene-1-ol

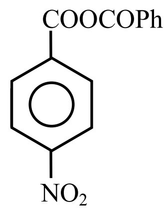
30. 

Compound M and corresponding reagents is/are :

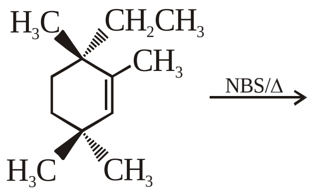
(A)  , $SOCl_2$, followed by base

(B)  , $NaBH_4$, followed by dil acid.

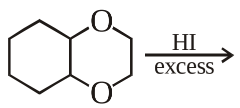
(C)  , Pyridine

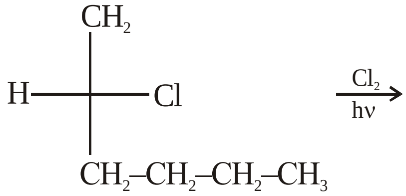
(D)  , PCl_5 , followed by pyridine

31. How many different products (including stereoisomers) are obtained in the following reaction ?



32. Calculate number of moles of HI involved in the following reaction :-

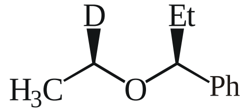


33.  (x). (x) = total number of di-chloro products :-

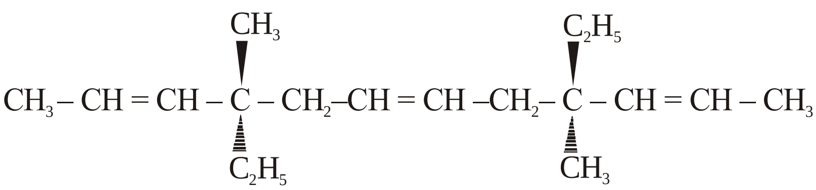
S-2-chloro hexane

34. Lithium di(3-pentyl) cuperate on reaction with an alkyl bromide produces (X) C_6H_{14}.The mixture of all monochloro isomers formed from (X) are subjected to fractional distillation. Find the number of fractions obtained.:-

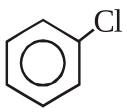
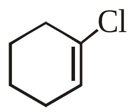
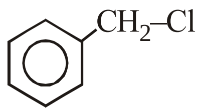
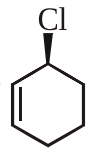
35. How many alkenes are possible on reaction of Cis-1-Chloro-1,4-diethyl cyclohexane with alcoholic caustic potash?

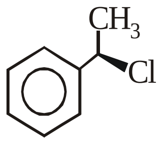
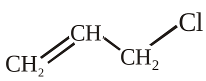
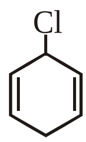
36. Number of major organic products obtained on hydrolysis of 

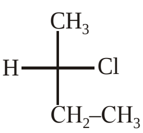
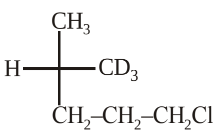
37. The number of optically active products obtained from the complete ozonolysis of the given compound is :

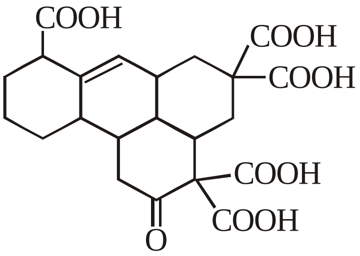


38. Number of compounds which slowly racemises on addition of $SbCl_5$.

(i)  (ii)  (iii)  (iv) 

(v)  (vi)  (vii)  (viii) $CH_2{=}CH{-}Cl$

(ix)  (x) 

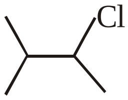
39.  $\xrightarrow{\Delta}$ Number of CO_2 evolved.

40. Treatment of papaverine ($C_{20}H_{21}O_4N$) with hot concentrated hydriodic acid yields CH_3I, indicating the presence of the methoxyl group $-OCH_3$. When 4.24 mg of papaverine is treated with hydriodic acid and the CH_3I thus formed is passed into alcoholic silver nitrate, 11.62 mg of silver iodide is obtained.
How many methoxyl groups per molecule of papaverine would be present ?

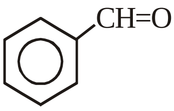
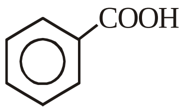
41. Identify total number of alkene products formed if 3-Chloro-3-methyl pentane react with alcoholic KOH and heat ?

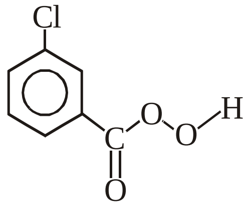
42. Number of reactions in which unsaturated hydrocarbon is obtained as a major product :

(a) $Al_4C_3 \xrightarrow{H_2O}$ (b) $R-C\equiv C-R \xrightarrow[\text{liq.}NH_3]{Na}$ (c) $Be_2C \xrightarrow{H_2O}$

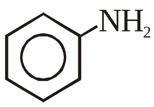
(d)  $\xrightarrow[\Delta]{MeONa}$ (e) $CaC_2 \xrightarrow{H_2O}$ (f) $R-C\equiv C-R \xrightarrow[BaSO_4]{H_2Pd}$

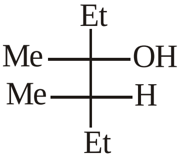
(g) $Mg_2C_3 \xrightarrow{H_2O}$

43. Number of reagents by which  can be converted in to  followed by acidification.

(i) $[Aq.(NH_3)_2]OH$ (ii)  (iii) $K_2Cr_2O_7/H^+, \Delta$ (iv) Conc. ^-OH

(v) $KMnO_4, \Delta$ (vi) $CuSO_4 + NH_4OH$ (vii) Zn-Hg/HCl (viii) Red P / HI

(ix) Alcoholic KCN (x) 

44.  $\xrightarrow{\text{conc.}H_2SO_4}$ Products

Optically pure

(a) Number of alkenes in the product mixture which can show only geometrical isomerism is [X]
(b) Number of alkenes in the product mixture which can show only optical isomerism is [Y]
(c) Number of alkenes present in the product mixture is [Z]
Represent your answer as XYZ

45. $$\underset{\substack{(P)\\ \text{dioic acid}\\ \text{(Shows geometrical isomerism)}}}{C_4H_4O_4} \xrightarrow[\text{catalyst}]{D_2 + \text{Wilkinson's}} \underset{(Y)}{C_4H_4D_2O_4}$$

Calculate total number of organic product (Y)

46. Calculate total number of all possible products when 1, 2 diphenylethene is reacted with chlorine water solution.

47. A hydrocarbon was found to have a molecular weight of 80-85. A 10.2 mg sample took up 8.40 ml of H_2 gas measured at 0ºC and 760 mm pressure. Ozonolysis yield only HCHO and OHC–CHO (Glyoxal).

If DU of hydrocarbon is x and total carbon present is y then find value of x + y.

48. Cyclohexanol + $K_2Cr_2O_7 + H^+ \longrightarrow [P] \xrightarrow{PhMgBr} \xrightarrow{H_2O} [Q] \xrightarrow{H^+ + O} [R] \xrightarrow{KMnO_4 \text{ dil}} [S]$

How many hydroxyl group will be present in the final product (S).

49. Consider the reaction sequence from M to T shown below. The overall yield of major product T from M is 60%. What is the amount of T obtained from 5.4 ml of M ?

[Use density of P = 1 gm ml^{-1}, Molar mass of C = 12.0, H = 1.0, O = 16.0 g mol^{-1})

$$\underset{\textbf{[M]}}{\text{But-1-yne}} \xrightarrow{NaNH_2} [P] \xrightarrow{\text{1-Bromo-2-chloroethane}} [Q] \xrightarrow[\substack{2.\ CH_3CH=O\\ 3.\ H^+}]{1.\ Mg+THF} [R]$$

$$[R] \xrightarrow{H_2 + Pd + CaCO_3} [S] \xrightarrow{CrO_3} [T]$$

50. cis & trans-2-bromocyclohexanol $\xrightarrow{HBr}$ Products

Calculate total number of major products in the reaction mixture.

Question Stem-1

For the following reaction scheme, percentage yields are given along the arrow.

$$\underset{\text{(1-Bromodecane)}}{n-C_{10}H_{21}Br} + HC \equiv CNa \longrightarrow \underset{(83\,g)}{[P]}$$

$$P + \text{n-BuLi} \xrightarrow[100\%]{} [Q]$$

$$\text{5-Methylhex-1-ene} \xrightarrow[100\%]{HBr + H_2O_2} [R]$$

$$[Q] + [R] \xrightarrow{60\%} \underset{(x\ g)}{[S]}$$

$$[S] + H_2 \xrightarrow[100\%]{CaCO_3/Pd} [T]$$

$$\underset{}{[T]} \xrightarrow[80\%]{\text{Peroxybenzoic acid}} \underset{(y\,g)}{[U]}$$

x g and y g are mass of S & U respectively.

(Use : Molar mass (in g mol^{-1}) of H,C, O and Br as 1, 12, 16 and 80 respectively).

51. The value of x is______

52. The value of y is______

Question Stem-2

For the following reaction scheme, percentage yields are given along the arrow.

$$\underset{\substack{\text{1,2,3-Tribromopropane} \\ (56.2g)}}{[P]} \xrightarrow[40\%]{\text{alc KOH}} [Q] \xrightarrow[100\%]{\text{aq. NaOH}} [R] \xrightarrow[60\%]{\text{alc.KOH}/\Delta} \underset{\substack{(x\,g) \\ (\text{Alkyne})}}{[S]}$$

$$[S] \xrightarrow[100\%]{H^+ + \Delta} [T] \xrightarrow[2.H^+, \quad 40\%]{1.\ CH_2MgBr + THF} \underset{(y\,g)}{[U]}$$

x g and y g are mass of S and U respectively.

(Use : Molar mass (in g mol^{-1}) of H,C, O and Br as 1, 12, 16 and 80 respectively).

53. The value of x is______

54. The value of y is______

Answer key

1.	(B)	**2.**	**(C)**	**3.**	(B)	**4.**	(C)	**5.**	(C)
6.	(B)	**7.**	(C)	**8.**	(A)	**9.**	(D)	**10.**	(C)
11.	(A, B, D)	**12.**	(B, D)	**13.**	(A, C)	**14.**	(A, C, D)	**15.**	(A,B,C)
16.	(A, C, D)	**17.**	(A,B,C)	**18.**	(A,B,C)	**19.**	(A,B,D)	**20.**	(B,C)
21.	(A,C)	**22.**	(A,B,D)	**23.**	(A,B,C)	**24.**	(A,D)	**25.**	(C,D)
26.	(B,D)	**27.**	(A,B,D)	**28.**	(A,C)	**29.**	(C,D)	**30.**	(A,C,D)
31.	3	**32.**	6	**33.**	9	**34.**	5	**35.**	4
36.	3	**37.**	1	**38.**	4	**39.**	4	**40.**	4
41.	3	**42.**	5	**43.**	5	**44.**	115	**45.**	3
46.	7	**47.**	9 (3 + 6)	**48.**	2	**49.**	7.56 g	**50.**	2
51.	79.2 gm	**52.**	67.70 gm	**53.**	2.688 gm	**54.**	1.42 gm		

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